(e) Let G be the group of quaternions of norm 1. Show that this group is isomorphic to $SU(2)$. (Thus geometrically $SU(2)$ is the 3-dimensional sphere.)

(f) Consider the action of G on the space $V \subset \mathbb{H}$ spanned by i, j, k, by $x \to qxq^{-1}$, $q \in G$, $x \in V$. Since this action preserves the norm on V, we have a homomorphism $h : SU(2) \to SO(3)$, where $SO(3)$ is the group of rotations of the 3-dimensional Euclidean space. Show that this homomorphism is surjective and that its kernel is $\{1, -1\}$.

Problem 4.12.8. It is known that the classification of finite subgroups of $SO(3)$ is as follows:

1) the cyclic group $\mathbb{Z}/n\mathbb{Z}$, $n \geq 1$, generated by a rotation by $2\pi/n$ around an axis;

2) the dihedral group D_n of order $2n$, $n \geq 2$ (the group of rotational symmetries in 3-space of a plane containing a regular n-gon[2];

3) the group of rotations of a regular tetrahedron (A_4);

4) the group of rotations of a cube or regular octahedron (S_4);

5) the group of rotations of a regular dodecahedron or icosahedron (A_5).

(a) Derive this classification.

Hint: Let G be a finite subgroup of $SO(3)$. Consider the action of G on the unit sphere. A point of the sphere preserved by some nontrivial element of G is called a pole. Show that every nontrivial element of G fixes a unique pair of opposite poles and that the subgroup of G fixing a particular pole P is cyclic, of some order m (called the order of P). Thus the orbit of P has n/m elements, where $n = |G|$. Now let $P_1, \ldots, P_k$ be a collection of poles representing all the orbits of G on the set of poles (one representative per orbit), and let $m_1, \ldots, m_k$ be their orders. By counting nontrivial elements of G, show that

$$2\left(1 - \frac{1}{n}\right) = \sum_i \left(1 - \frac{1}{m_i}\right).$$

Then find all possible m_i and n that can satisfy this equation and classify the corresponding groups.

[2] A regular 2-gon is just a line segment.